

Syarfandi "Andi" Achmad

Data Scientist | ML Engineering & Predictive Analytics Specialist Makassar, Indonesia | +62 852-5652-2335 | syarfandi.achmad@gmail.com | Portfolio: syarfandi.github.io | LinkedIn

Professional Summary

Insightful **Data Scientist** with 4+ years extracting actionable intelligence from complex datasets. Expert in **Python**, **Machine Learning**, **Statistical Modeling**, and **Predictive Analytics** using real-world production data. Proven track record delivering data-driven solutions for government (9M+ citizen records) and enterprise retail (100+ module POS). Skilled at building ML pipelines, interactive dashboards, and AI-powered applications.

Core Competencies

- ML/AI:** Scikit-Learn, TensorFlow, ARIMA/Prophet, K-Means, DBSCAN, NLP, LLM Integration (Gemini AI), Anomaly Detection.
- Python:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Jupyter Notebooks, Feature Engineering.
- SQL & Data:** PostgreSQL (Stored Procedures, Aggregation), Supabase, ETL Pipelines, Data Warehousing, Data Cleaning.
- Visualization:** Matplotlib, Seaborn, D3.js, Interactive Dashboards, Geospatial Visualization.
- Statistics:** Hypothesis Testing, Regression, Classification, A/B Testing, Time-Series Analysis, Bayesian Methods.
- Tools:** Git, Docker, GitHub Actions, Jupyter, Google Colab, Linux.

Projects

- Kawan Sedarah (Blood Stock):** A **command center** dashboard that tracks **blood stock** availability across donor units in **real-time**, powered by **Supabase** as its core backend.
- Portal Data Sulse:** A unified information system delivering **data visualization**, **monitoring**, and religious **statistical analysis** across all districts in South Sulawesi in **real-time**.
- AI Virtual Assistant:** A **Conversational AI**-powered virtual assistant featuring **Realtime Lipsync** on 3D avatars, backed by **Voice AI** and **LLM** for immersive conversational interactions.
- GetKasir Enterprise POS:** A large-scale **Enterprise** POS ecosystem with **100+ modules**, integrating **AI-driven** sales prediction, **biometric** authentication, and a scalable **cloud-native** architecture.
- PPID Kemenag Kota Parepare:** A public information disclosure portal for **PPID Kemenag** providing access to **official documents**, **performance reports**, and an intelligent chatbot powered by **Google Gemini AI**.
- OP Tracker:** A premium logbook for tracking **One Piece** watch progress without login — equipped with automatic **filler filter**, **saga & arc** navigation map, and a gamified **pirate rank** system.
- QR Scanner Presence App:** A modern **attendance solution** leveraging **QR Code** scanning for instant participant check-in, built on a **Next.js** + **Supabase** stack with **real-time** sync.
- Malmora (Reseller & Dropship):** A **reseller & dropship** marketplace tailored for Muslim lifestyle products, enabling anyone to sell without physical stock across **Sulawesi** and **Eastern Indonesia**.
- Email Server Makassar Gov:** A city-scale **Mailcow**-based email cluster serving **200+ official domains** for Makassar City Government, hardened with **SPF/DKIM/DMARC** security protocols and **high-availability** setup.
- Helpdesk Kota Makassar:** An **e-government** system for handling **citizen complaints** digitally — featuring **real-time ticket tracking**, automated escalation, and cross-department integration across **Makassar City agencies**.
- Virtual Museum Makassar:** An immersive digital museum experience with **360° panoramic** tours, interactive **3D artifact** collections, and **audio-visual** narratives preserving South Sulawesi cultural heritage.
- AI-Haram VR:** A high-precision **virtual reality** simulation of **AI-Masjid Al-Haram** and **AI-Masjid An-Nabawi**, developed in **Unity** and optimized for both **VR headsets** and mobile devices as a Hajj & Umrh educational tool.
- Paralluta (Lalu Lintas):** A smart platform for **traffic management** in Makassar City — covering **road condition monitoring**, violation logging, and centralized **field officer coordination**.
- SOP Pelaporan Insiden Siber:** A **SOP** framework and **cyber incident response protocols** designed for **Makassar City Government** IT infrastructure, aligned with **BSSN** and **NIST Cybersecurity Framework** standards.

Experience

Computer Specialist (Civil Servant) | Ministry of Religious Affairs

2025 – Present | Makassar

- Built** the **Data Sulse** analytics engine — PostgreSQL aggregation pipelines, statistical dashboards, and trend analysis for 9M+ citizen service records.
- Developed** predictive models for resource allocation, using historical data to forecast demand across 24 regencies.
- Integrated Google Gemini AI** into public portals (PPID), implementing NLP-powered document search and automated response generation.

DevOps & Infrastructure Expert | Diskominfo Makassar

2023 – 2025 | Makassar

- Built** sentiment analysis models for **Helpdesk Makassar** complaint data — NLP classification to auto-prioritize critical urban issues.
- Developed** anomaly detection algorithms for government email clusters, identifying patterns across 200+ domains.

Co-Founder | GetKasir (Enterprise POS)

2019 – Present | Remote

- Built** sales forecasting models using time-series (ARIMA/Prophet) for **100+ module POS**, improving inventory accuracy by 40%.
- Developed** merchant clustering (K-Means/DBSCAN) for churn risk identification and targeted marketing segments.
- Designed** automated ETL pipelines for high-volume transaction data — cleaning, feature engineering, aggregation.

SQA & DevOps Engineer | Malmora (Startup)

2020 – 2023 | Remote

- Analyzed** sales data and purchase patterns to optimize product ranking and recommendation algorithms.
- Built** A/B testing frameworks for conversion optimization, analyzing statistical significance on order completion.
- Developed** regional market dashboards tracking seller performance, demand trends, and logistics metrics.

Awards

- National Cyber Exercise #10** – BSSN (2024)
- Top 10** – Samsung Next Apps 4.0 (2017)
- Microsoft CityApp** – Participant

Education

- Bachelor of Computer Science (S.Kom)** | UIN Alauddin Makassar *GPA: 3.22 / 4.00*

Data Scientist Expertise Keywords

Python, SQL, Machine Learning, Deep Learning, Scikit-Learn, TensorFlow, PyTorch, Pandas, NumPy, NLP, LLMs, Computer Vision, Predictive Modeling, Statistics, Time Series Forecasting, A/B Testing, Feature Engineering, Model Deployment, Jupyter, MLOps.

Future-Ready Technical Roadmap

To stay competitive in the global Data & Analytics market, I am actively implementing:

- Predictive Analytics:** Implementing advanced ML models for forecasting and trend analysis.
- Real-time Streaming:** Building low-latency data pipelines using Apache Flink and Spark Streaming.
- Vector Databases for LLMs:** Architecting data storage for large language model retrieval (RAG).
- Automated Data Governance:** Ensuring data compliance and quality through automated AI-driven auditing.